

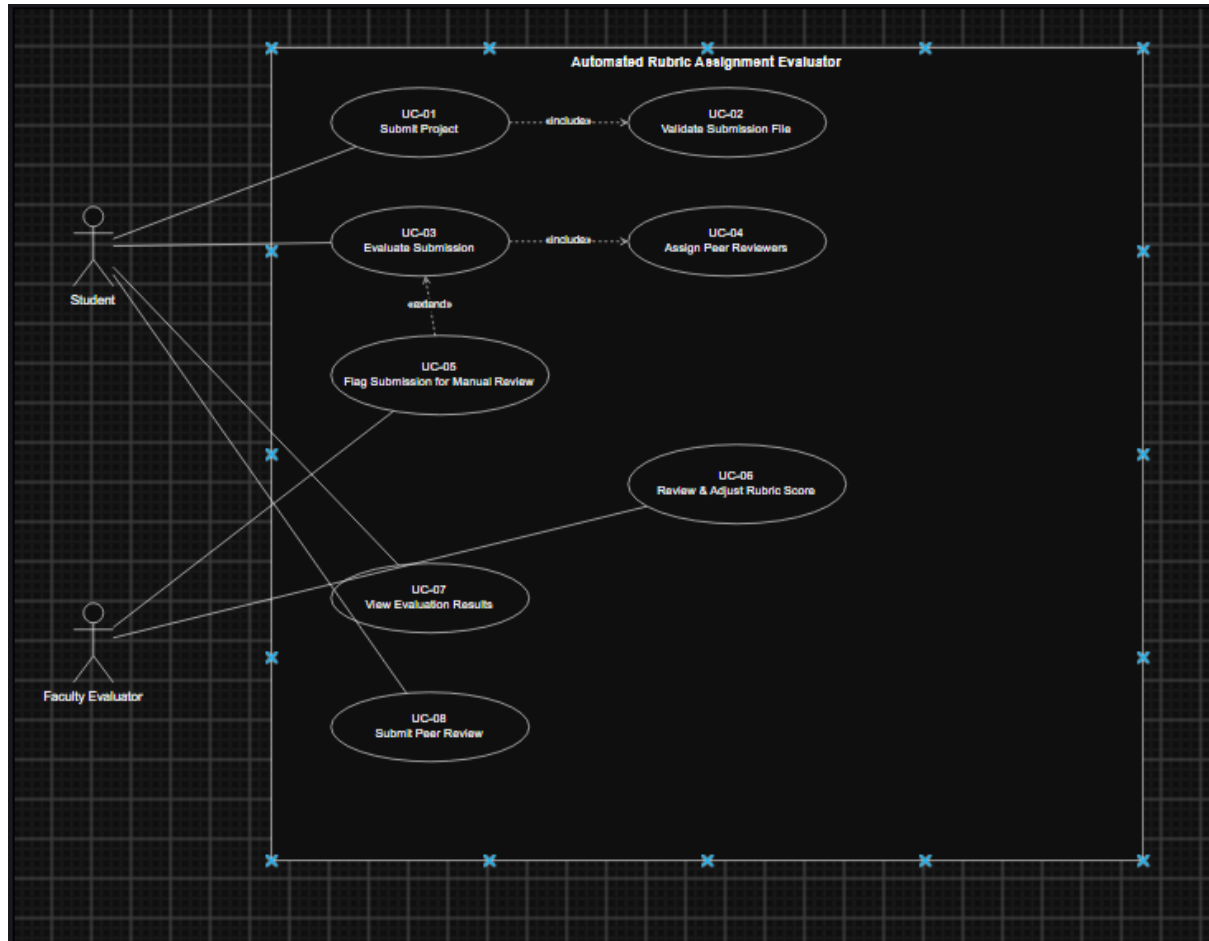
Problem Statement: Automated Rubric Assignment Evaluator

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Target Stakeholders / Actors: Student, Faculty Evaluator

UML diagram:



Requirement Table:

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system should accept student project ZIP files, run the given test cases, and generate a rubric-based score.	High	Pass: Tests are run and the score is displayed within 10 seconds. Fail: The submission is not processed or the score is incorrect.	This is the main function of the system.

FR-002	Functional	The system should automatically assign each evaluated submission to the required number of peer reviewers.	High	Pass: Each submission is assigned to the required reviewers within 5 minutes and no reviewer gets more than the allowed number of submissions. Fail: A submission is left unassigned or a reviewer gets too many submissions.	Reduces the need for the Faculty Evaluator to manually assign reviews.
FR-003	Functional	The system should allow the Faculty Evaluator to view, change, and finalize the score given by the automated evaluation.	High	Pass: The changed score is saved, the total is updated, and the change is recorded. Fail: The change is not saved or the total is not updated.	The Faculty Evaluator should have the final control over the grade.
FR-004	Functional	The system should show the final score to the Student and notify them after the Faculty Evaluator completes the evaluation.	Medium	Pass: The Student can see the final score and receives a notification within 2 minutes. Fail: The score or notification is missing.	Students need to know their final evaluation results.
FR-005	Functional	The system should check the uploaded file for the correct format, size, and required files before accepting it.	Medium	Pass: Invalid files are rejected with an error message and the Student can upload another file. Fail: An invalid file is accepted or no useful error is shown.	Prevents invalid submissions from entering the evaluation process.

NFR-001	Performance & Security	The system should support up to 100 students submitting projects at the same time without losing submissions or using more than 1.5 GB of memory.	High	Pass: 100 submissions are processed without losing any submission and memory stays below 1.5 GB. Fail: Submissions are lost or memory goes above the limit.	The system may receive many submissions around deadlines.
NFR-002	Reliability	Every accepted submission should end with either an evaluated score or a flagged status. The system should maintain at least 99.9% reliability.	High	Pass: At least 999 out of 1000 test submissions reach a final status and none disappear from the system. Fail: A submission is lost or more than one does not reach a final status.	Makes sure submitted projects are not lost during evaluation.

Actors:

- Student
- Faculty Evaluator

Use Cases:

- UC-01 Submit Project
- UC-02 Validate Submission File
- UC-03 Evaluate Submission
- UC-04 Assign Peer Reviewers
- UC-05 Flag Submission for Manual Review
- UC-06 Review & Adjust Rubric Score
- UC-07 View Evaluation Results
- UC-08 Submit Peer Review

Student is connected to:

- UC-01 Submit Project
- UC-03 Evaluate Submission
- UC-07 View Evaluation Results
- UC-08 Submit Peer Review

Faculty Evaluator is connected to:

- UC-05 Flag Submission for Manual Review

- UC-06 Review & Adjust Rubric Score

Include relationships:

- UC-01 Submit Project → «include» → UC-02 Validate Submission File
- UC-03 Evaluate Submission → «include» → UC-04 Assign Peer Reviewers

Extend relationship:

- UC-05 Flag Submission for Manual Review → «extend» → UC-03 Evaluate Submission

The **Flag Submission** use case happens only when automated evaluation fails, such as when the code does not compile, the sandbox times out, or there are too many errors.

Use-Case Flow : UC-03 Evaluate Submission

Primary Actor: Student

Includes: UC-04 Assign Peer Reviewers

Extended By: UC-05 Flag Submission for Manual Review

Preconditions

1. The Student is logged in and is enrolled in the assignment.
2. The submitted file has passed the file validation check.
3. The assignment has a rubric and test cases configured.

Postconditions

Success:

- The rubric score is calculated and saved.
- Submission status becomes **"Evaluated :Pending Faculty Review"**.
- Peer reviewers are assigned.

Failure:

- Submission status becomes **"Flagged Manual Review Required"**.
- The Faculty Evaluator is notified.

Main Flow

1. The Student's submission is added to the evaluation queue.
2. The system gets the rubric and test cases for the assignment.
3. The system runs the test cases on the submitted code in a sandbox.
4. The system checks the code for syntax errors.
5. The system calculates the score based on the test results.
6. The system displays the rubric-wise score within 10 seconds.

7. The system saves the score and changes the submission status to **"Evaluated"**.
8. The system automatically assigns peer reviewers.
9. The evaluation process is completed.

Alternate Flow : Automated Evaluation Fails

1. The system detects a compilation error, sandbox timeout, or too many errors.
2. Automated evaluation is stopped.
3. The submission is marked **"Flagged Manual Review Required"**.
4. The Faculty Evaluator is notified along with the error details.
5. The Faculty Evaluator reviews the submission and changes or assigns the score.
6. The submission moves to the manual review process.